Article Recommender System

DS300.Q11 - Final Project

Presented by:
T. Nhat[1] L. M. Nhut[1] N. H. Phat[1]
Instructor: Master. Huynh Van Tin[1]

[1]Department of Computer Science
University of Information Technology

DS300 Final Project Report

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2. Dataset Construction and EDA
3. State-of-the-Art (SOTA) Baselines
4. Proposed Method: MA-HGN
5. Engineering Implementation
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Motivation: Why News Recommendation?

In the digital era, news platforms face **Information Overload**:

- **User Experience:** Navigating thousands of daily articles is overwhelming.
- **Retention:** Personalized content increases user engagement and platform stickiness.
- **Monetization:** Better engagement leads to higher ad revenue and subscriptions.

Target: Build a system that understands both *Content semantics* and *User behavior*.

Research Objectives

Our goal is to develop a state-of-the-art recommendation engine for Vietnamese news:

- **Hybrid Logic:** Combine Collaborative Filtering (CF) with Content-Based (CB) techniques.
- **Graph Learning:** Leverage GNNs to model complex user-article interactions.
- **Multimodal Fusion:** Incorporate Social (replies) and Temporal (time-decay) signals.
- **Vietnamese NLP:** Optimize PhoBERT embeddings for news semantics.

Problem Statement

Challenge: How to recommend relevant news articles to users?

Key Challenges:

1. **Cold Start:** New users/articles have no interaction history
2. **Sparse Data:** Most users read very few articles (99.9% sparsity)
3. **Vietnamese Language:** Limited NLP resources for Vietnamese text
4. **Dynamic Content:** News articles are time-sensitive

Our Approach: Graph-based Contrastive Learning with PhoBERT embeddings

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VnExpress Data Crawling

Data Source: VnExpress.net (largest Vietnamese news portal)

Categories Crawled:

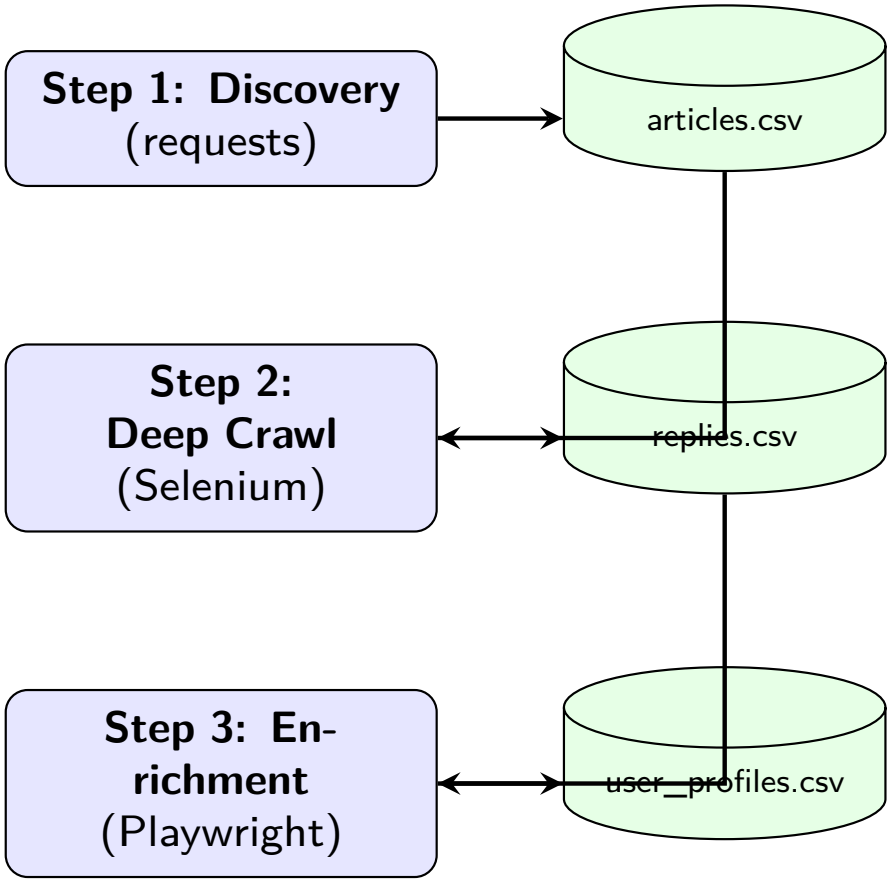
- 'Thế giới'
- 'Giáo dục'
- 'Thể thao'
- 'Thời sự'
- 'Kinh doanh'
- 'Khoa học công nghệ'

Collected Data:

- Articles: title, content, author, publish date
- Metadata: category, tags (per article)
- Comments: user ID, comment text, reactions, replies
- User profiles: user ID, username, join date

Data Collection Pipeline: 3-Step Workflow

Our crawler implements a resilient 3-stage architecture to handle high-volume news data:



Parallel scanning of category pages to find new URLs.

Parallel browser workers to extract deep comment threads & nested replies.

Scraping full history for all unique active commenters.

Pipeline Orchestration & Efficiency

- **Parallelism:** `Step 2` (Selenium) is resource-intensive. We split the workload across N parallel browser instances using `multiprocessing`.
- **Resiliency:**
  - **Caching:** Avoids redundant HTTP requests for static content.
  - **Checkpointing:** Atomic writes to CSV allow resuming from the last article in case of failure.
  - **Headless Execution:** Minimizes resource overhead in production environments.
- **Tool Stack:**
  - **Requests:** High-speed metadata extraction.
  - **Selenium:** Handling complex JS (Load More buttons).
  - **Playwright:** Modern, fast engine for user history scraping.

Dataset Overview: High-Level Statistics

Raw Behavioral Data:

- **3,685** Articles
- **42,050** Comments
- **6** News Categories
- **19,842** User Profiles

Graph Interaction Set (G1-G3):

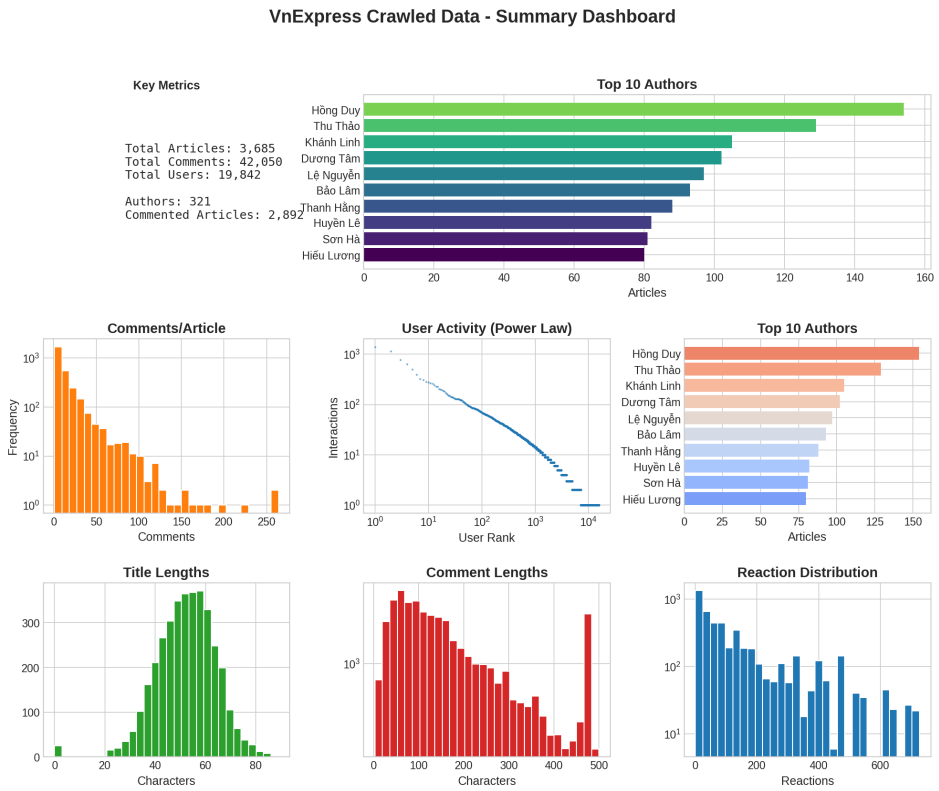
- **13,117** Unique Users
- **3,645** Active Articles
- **38,505** Total Interactions
- **Sparsity: 99.919%**

Dataset Overview (VnExpress)

Statistics:

- **Users:** 13,117
- **Articles:** 3,645
- **Interactions:** 38,505
- **Sparsity:** 99.92%

The dataset is highly sparse, typical of news domains where users read only a tiny fraction of daily content.



Data Sample: Interaction & Content

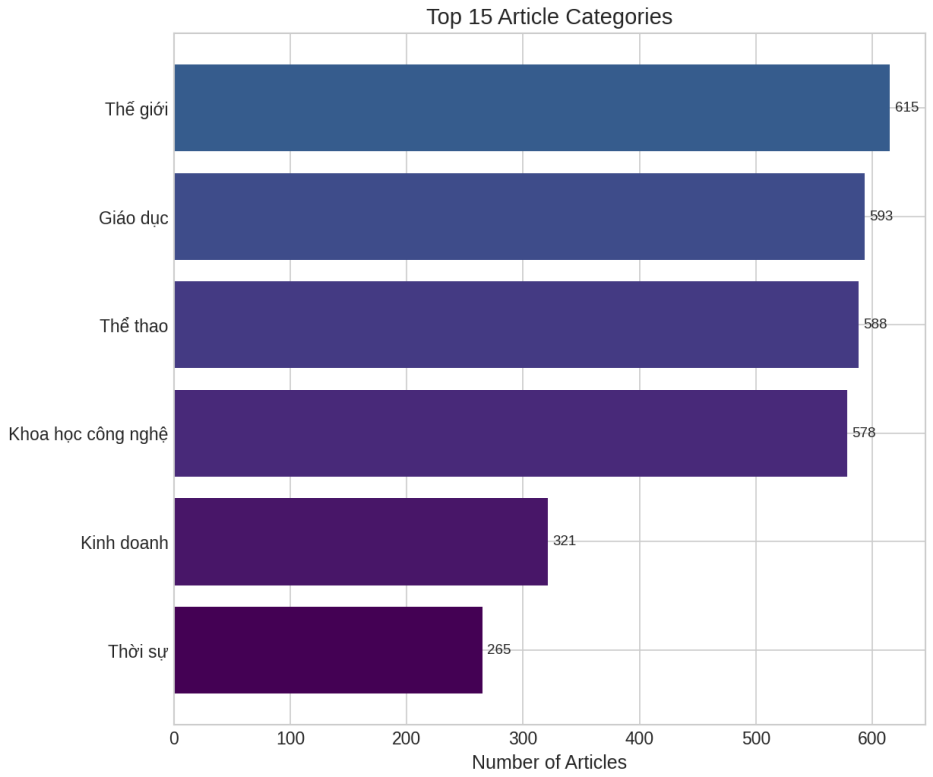
1. **Content Sample (Article)**
"Chinh phục loạt 'ông lớn' công nghệ sau hơn 500 lần bị từ chối"
→ **Tags:** [Meta, Amazon, Apple, Tesla, kĩ sư công nghệ]

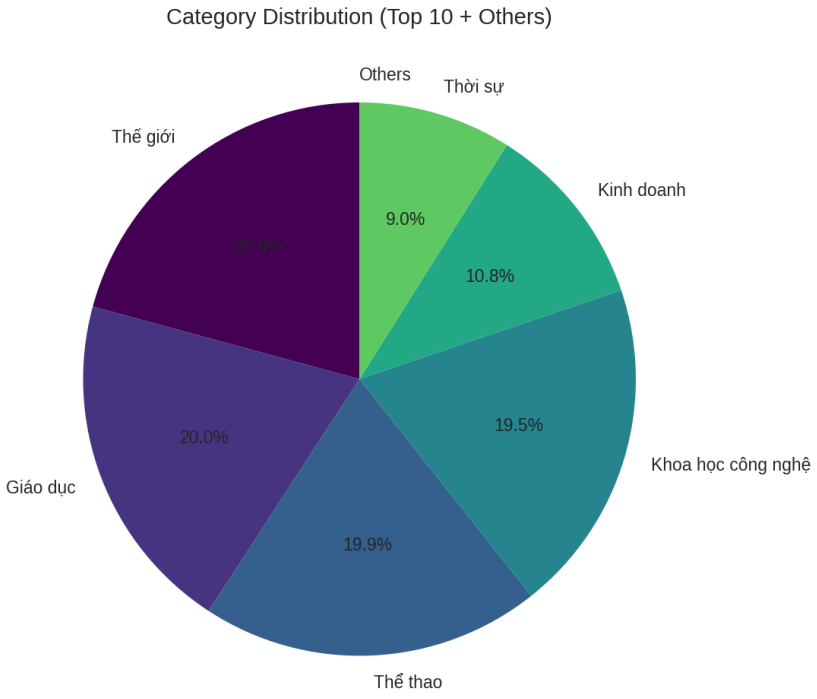
2. **Social Sample (Comment/Reply)**
   - **User A:** "Nếu giỏi thì cố gắng mua lại mấy công ty này..."
   - **User B:** (Reply to A) "Mua lại Meta chắc cần nghìn tỷ USD..."
   ⇒ Social Connection detected

Feature Mapping

- **Article** → PhoBERT Embedding ($d = 64$) → Node Feature.
- **User** → Interaction aggregation → Node Embedding.
- **Reply** → User-User Edge in the Heterogeneous Graph.

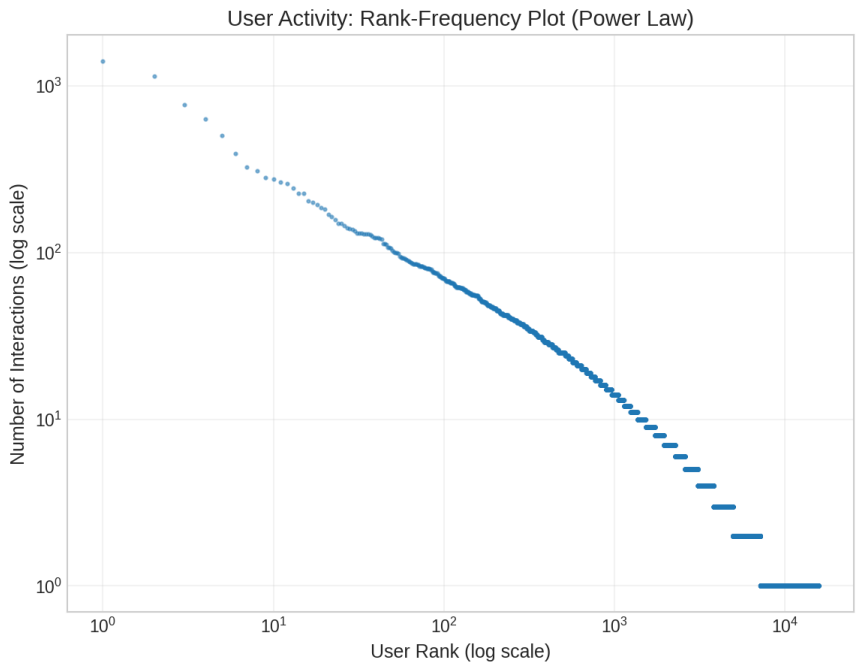
Category Distribution (VnExpress)

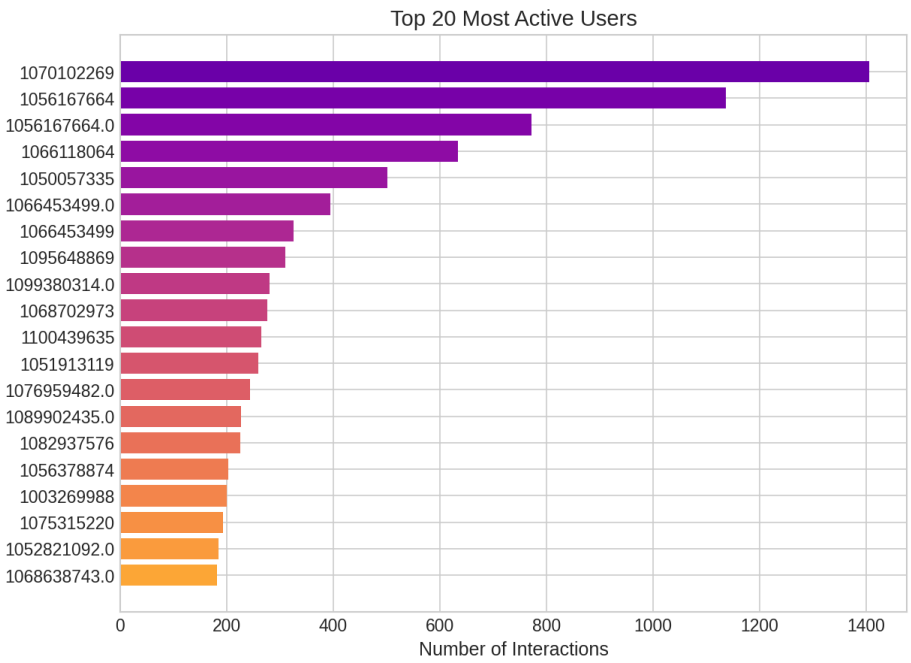




- **Thế giới** and **Thời sự** are the dominant categories.
- Skewed distribution reflects editorial priorities.

User Engagement Patterns (1/2)

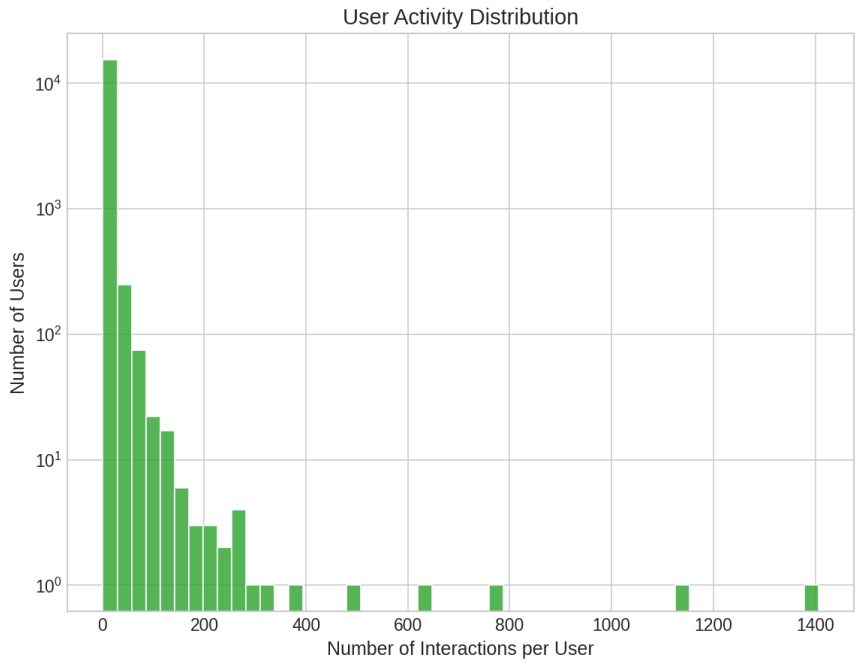


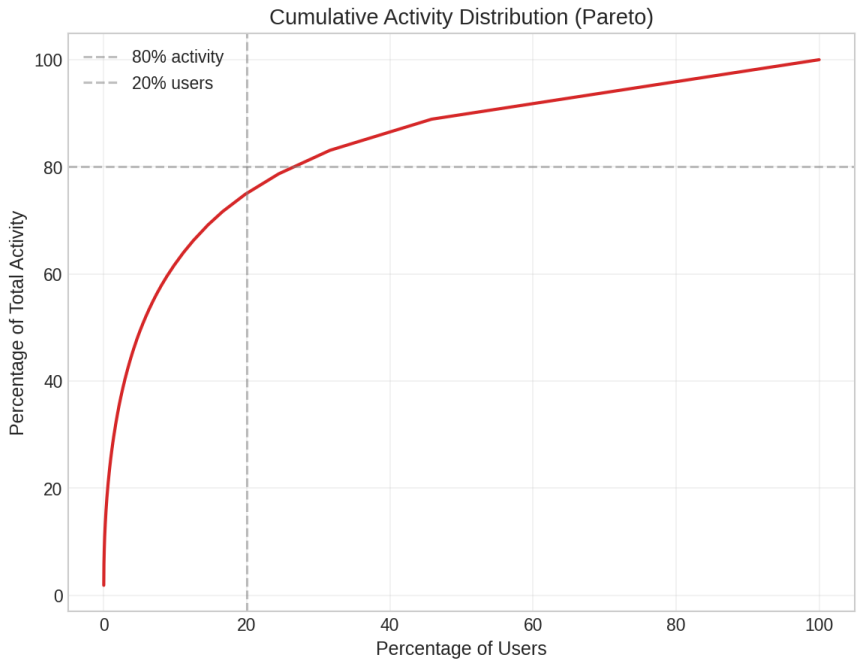


Distribution Analysis

Article popularity follows Zipf's law. Top 10 users account for 15% of all interactions.

User Engagement Patterns (2/2)

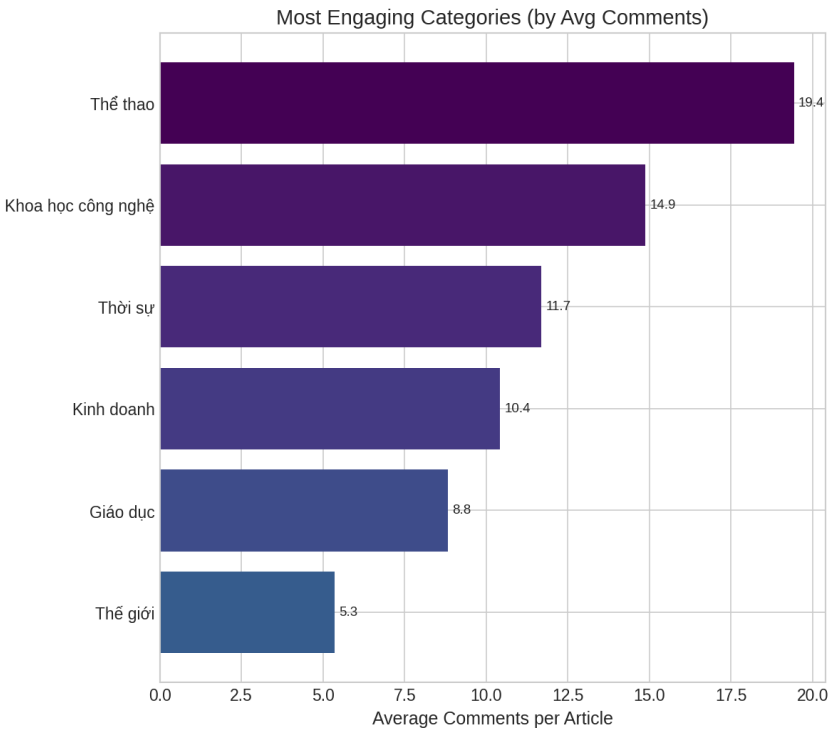


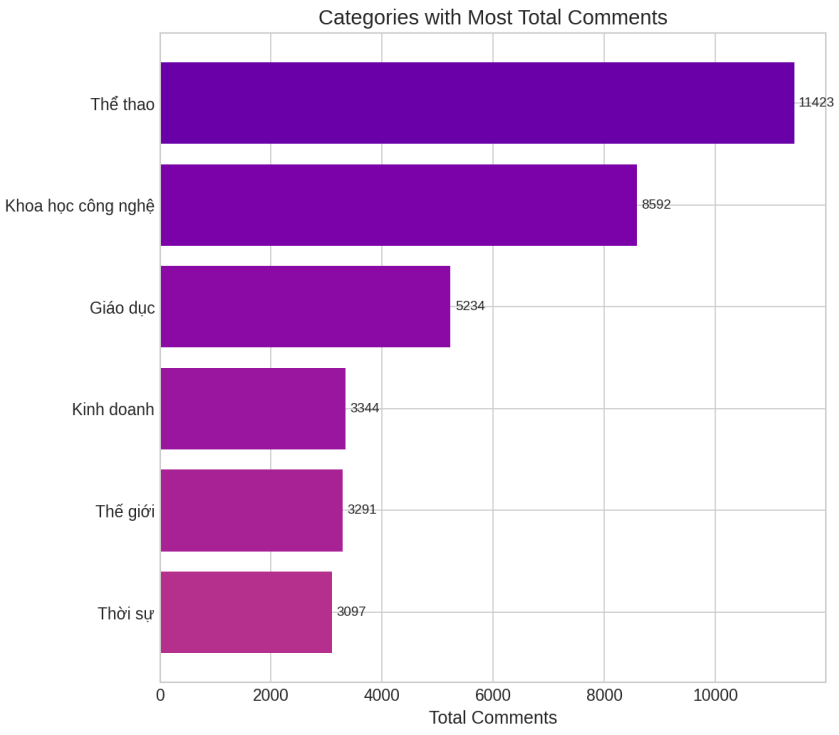


Power Law Observation

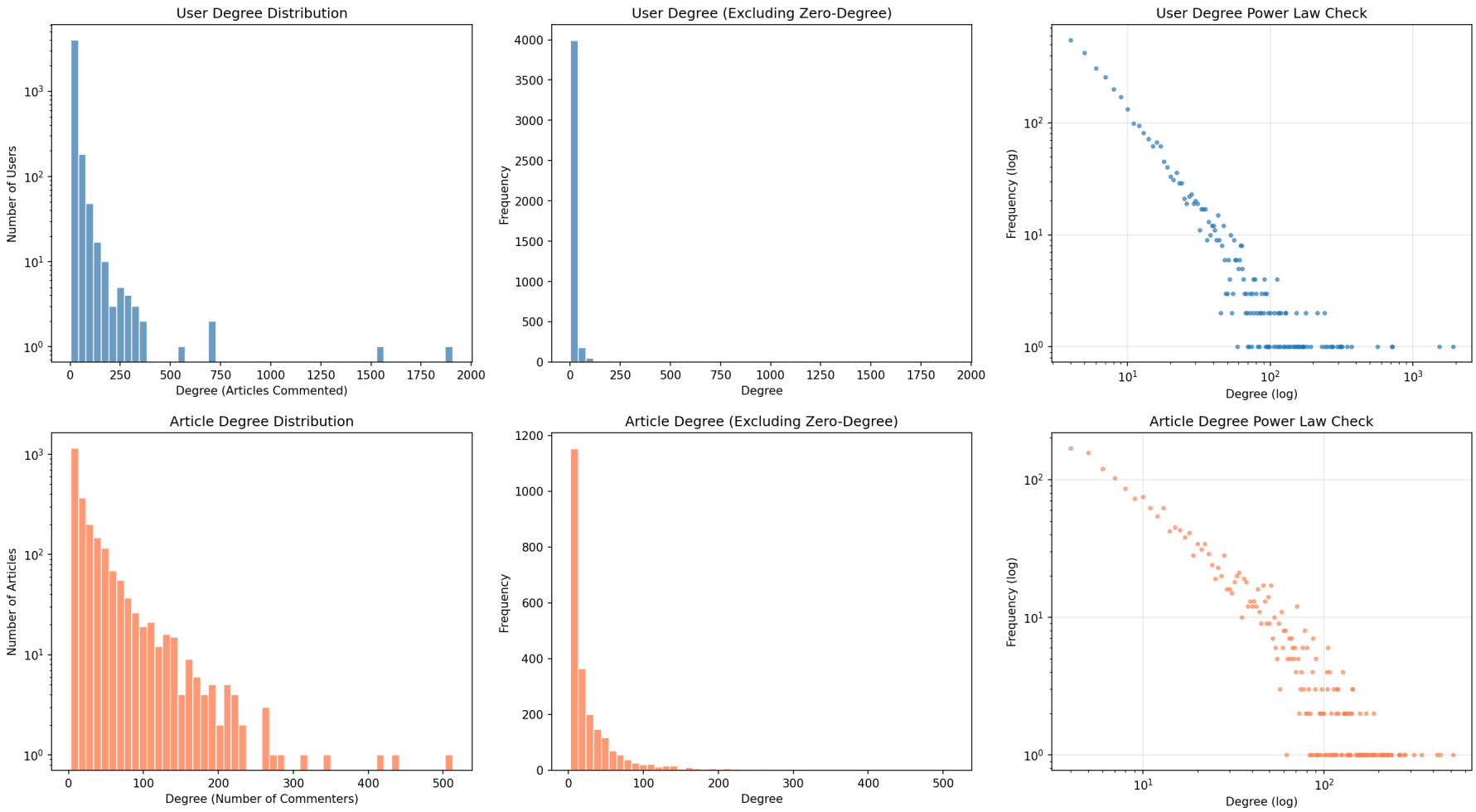
Long-tail distribution: 80% of users have fewer than 10 interactions.

Engagement by Category



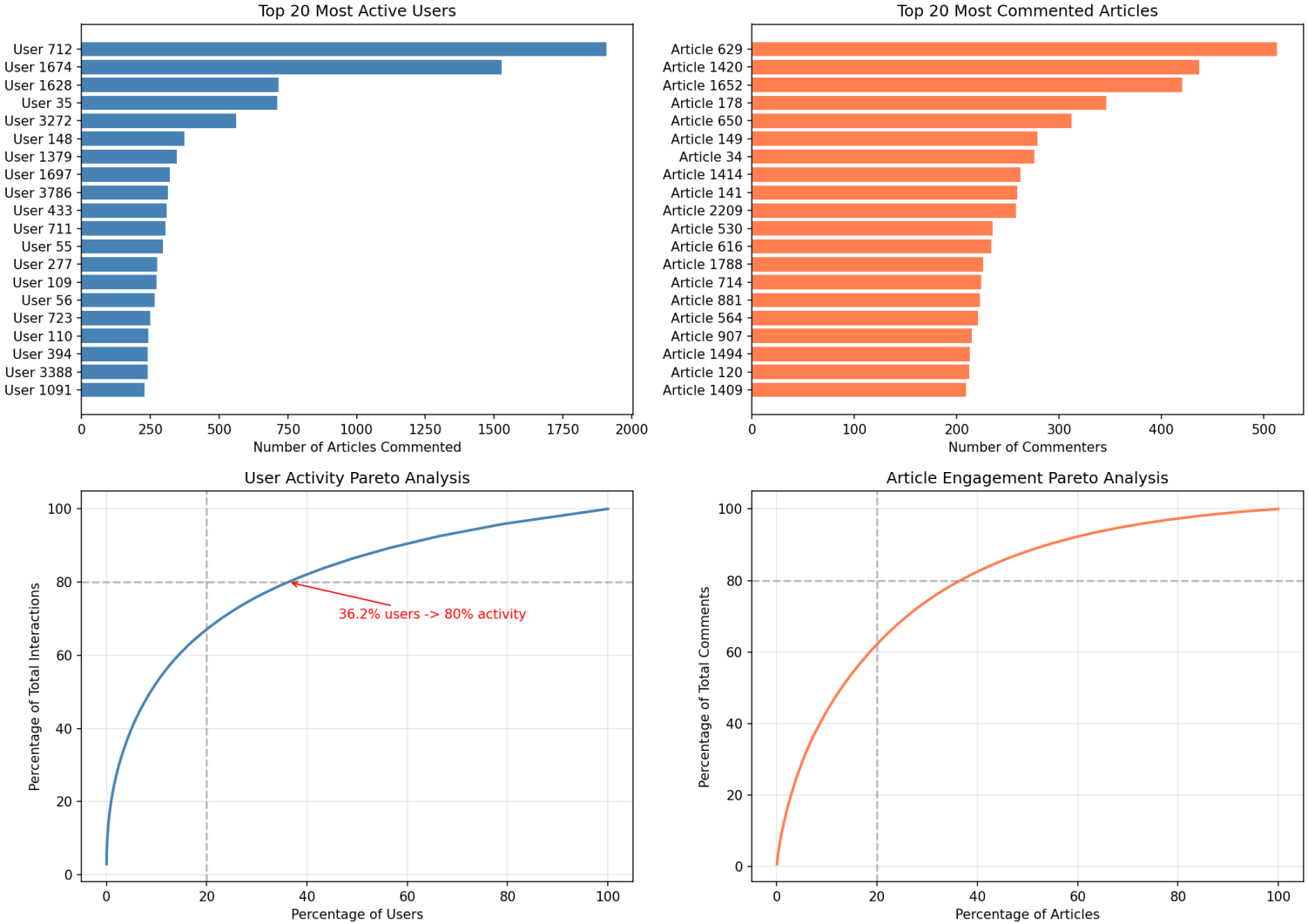


Graph Degree Distribution

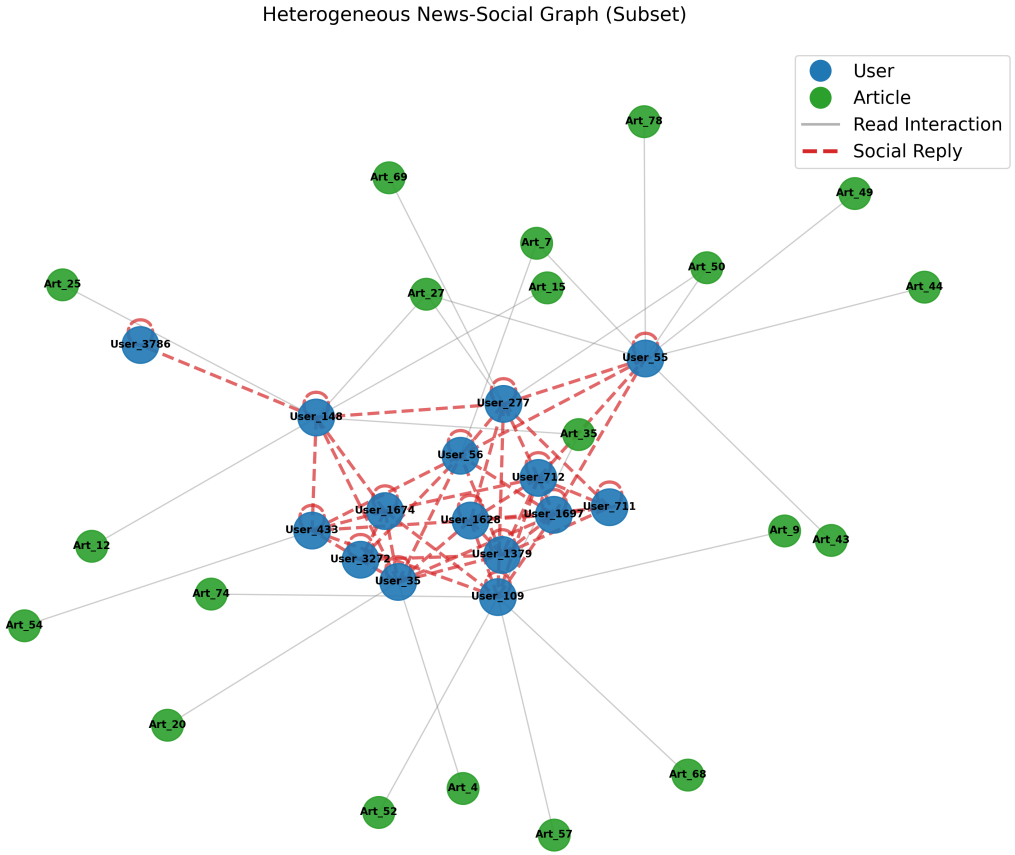


- User degrees peak at very low values (high cold-start risk).
- Article degrees are more distributed but still sparse.

Graph Connectivity & Density



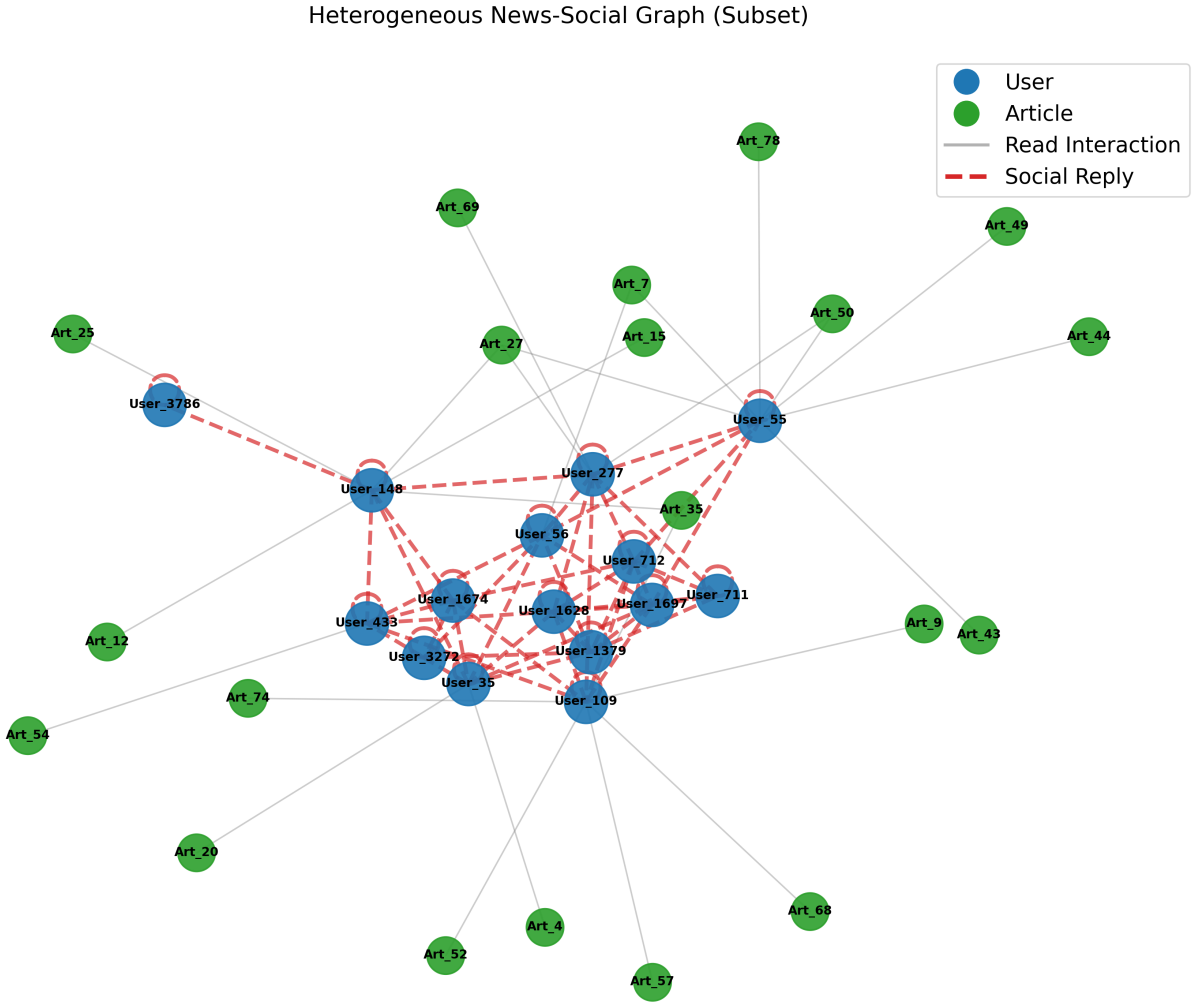
Graph Topology: A Heterogeneous Network



- **Blue Nodes:** Active Users
- **Green Nodes:** News Articles
- **Dashed Edges:** Social Reply interactions

Graph Topology

Visualizing the User-Article-Comment bipartite graph structure.



The Sparsity Challenge: Mathematical Proof

Definition:
Sparsity measures the lack of interactions in the potential link space.

The Formula

$$\text{Sparsity} = 1 - \frac{|\mathcal{E}_{obs}|}{|\mathcal{V}_u| \times |\mathcal{V}_i|}$$

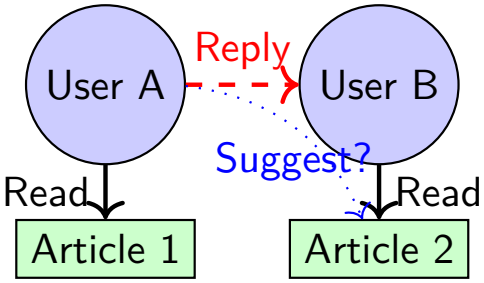
VnExpress Graph Calculation:

- Interactions ($|\mathcal{E}_{obs}|$): **38,505**
- Active Users ($|\mathcal{V}_u|$): **13,117**
- Active Articles ($|\mathcal{V}_i|$): **3,645**
- Total Possible Edges: $13,117 \times 3,645 = \mathbf{47,811,465}$

$$\Rightarrow \text{Sparsity} = 1 - \frac{38,505}{47,811,465} = \mathbf{99.919\%}$$

Core Innovation: Social Fusion

Idea: Use the "Reply Network" to bridge gaps between users.



The Logic:

- **Direct:** User A likes Article 1.
- **Social:** User A replies to User B $\rightarrow$ *Shared Interest*.
- **Inference:** User A is likely to enjoy Article 2 (read by B).

Impact on Sparsity

By adding User-User edges, the GNN can propagate "interests" even if User A has only read 1 article.

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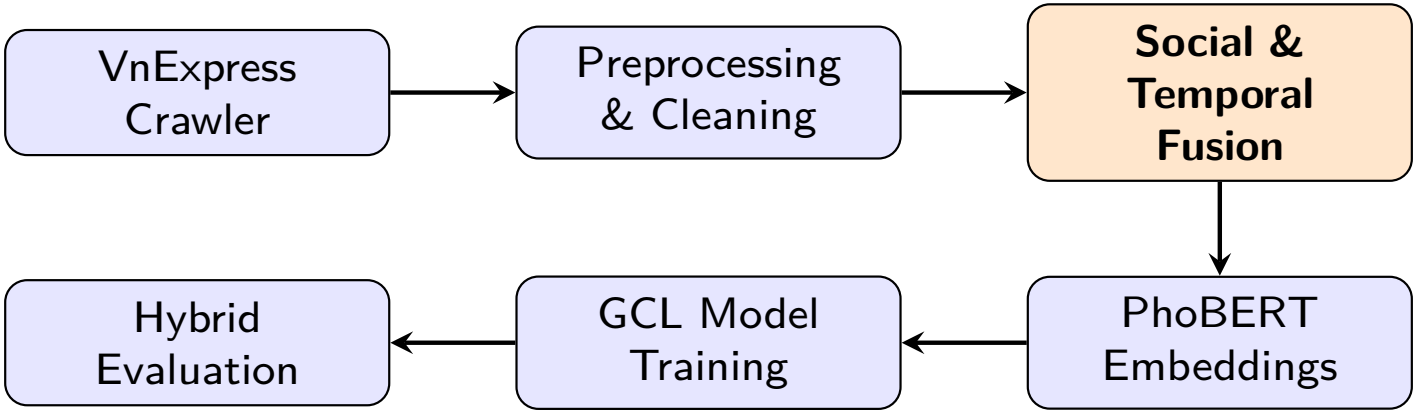
Current SOTA Approaches

We benchmark against three main categories of modern Recommender Systems:

1. **Graph Neural Networks (GNNs):**
   - **NGCF (2019):** First to encode collaborative signal into embeddings via graph propagation. High computational cost.
   - **LightGCN (2020):** Simplified GCN by removing non-linearities. Current industry standard for efficiency.
2. **Contrastive Learning (CL):**
   - **SimGCL (2022):** Augments graph structure with noise to learn robust representations. Handles sparsity better than pure GNNs.
3. **Content-Based & Hybrid:**
   - **PhoBERT:** State-of-the-art BERT model for Vietnamese, capturing deep semantic text features.

System Architecture



Pipeline: Crawl → Clean → Build Graph → Generate Embeddings → Train → Evaluate

Strategic Why: The Hybrid Synergy

Why combine **Content-Based** with **Graph Learning**?

- **Content-Based (NLP):**
  - Solves the **Cold-Start** problem (recommends new articles with 0 interactions).
  - Captures semantic similarity (PhoBERT).
- **Graph Learning (GNN):**
  - Captures **Collaborative Signals** (Users who like X also like Y).
  - Exploits **Social Signals** (Reply networks).

Hybrid Outcome

Content-Based provides "Freshness" while Graph Learning provides "Depth" and "Discovery".

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Inspiration & Theoretical Foundation

Where did this idea come from?
Our architecture is an adaptation of the **HGNR** framework, optimized with LightGCN mechanics.

1. HGNR (arXiv:2009.00799)

Heterogeneous GNN for Recommendation

- **Key Idea:** Users and items have multiple types of relationships (e.g., Click vs Reply).
- **Mechanism:** Use relation-specific aggregation steps.

2. LightGCN (He et al., SIGIR'20)

Simplified GCN for Recommendation

- **Key Idea:** Non-linearities (ReLU) are effectively redundant for collaborative filtering.
- **Mechanism:** Linear propagation + Layer averaging.

Motivation: Why a Heterogeneous GNN?

The Problem with Standard GNNs (LightGCN, NGCF):

- They treat the graph as **Homogeneous**: All nodes are just "IDs", all edges are just "links".
- **Failure Case:** A "Reply" (social trust) is treated identically to a "Click" (casual interest).

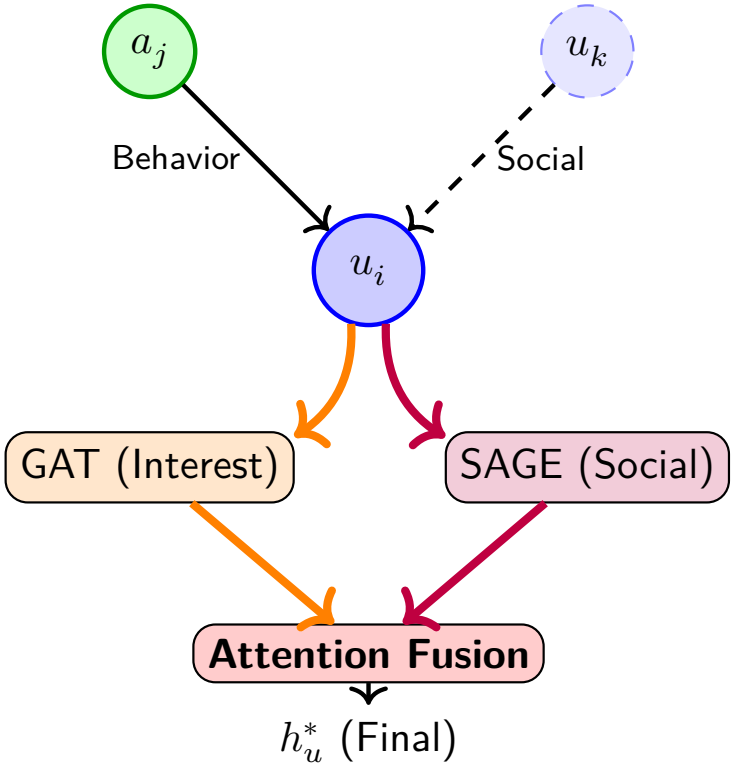
The VnExpress Reality:

- **Multiple Node Types:** Users $\neq$ Items.
- **Multiple Edge Types:**
  - `User-Click-Item`: Explicit Interest.
  - `User-Reply-User`: Social Influence / Trust.

The MA-HGN Solution

We need an architecture that aggregates **Social Signals** separately from **Content Signals** before fusing them.

MA-HGN: Multi-Aspect Heterogeneous GNN



Core Innovation: Dual-path Propagation

- **Path 1 (Interest):** history ($\mathcal{G}_{u-a}$) via **GATv2**.
- **Path 2 (Social):** friends ($\mathcal{G}_{u-u}$) via **GraphSAGE**.
- **Adaptive Fusion:** balances signal importance:

$$h_u = \alpha \cdot h_{int} + (1 - \alpha) \cdot h_{soc}$$

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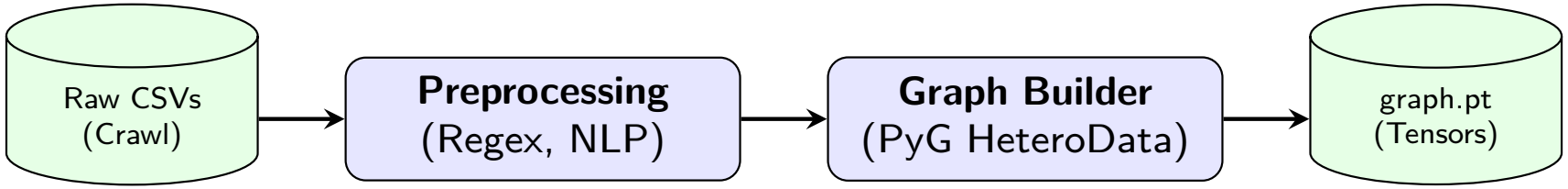
1. Data Engineering Pipeline (ETL)

From Raw HTML to Graph Tensors:



- **Text Processing:**
  - `vncorenlp` for segmentation.
  - `VN-SBERT` for 768d dense embeddings.
- **Graph Construction (G4):**
  - Map strings (User IDs) $\rightarrow$ Integer Indices $[0, N]$.
  - Create Edge Indices $(2 \times E)$ for 6 relation types.
  - Normalize adjacency matrices $\hat{A} = D^{-1/2} A D^{-1/2}$.

2. Model Training Pipeline

Orchestration Script: `train_cf_models.py`

Training Loop (Epoch t)

1. **Graph Encoder:** Pass graph $\mathcal{G}$ through L layers of LightGCN/SimGCL.
2. **Vector Retrieval:** Get User E_u and Item E_i embeddings.
3. **Negative Sampling:** Sample hard negatives (items user didn't interact with).
4. **Loss Computation:**

$$\mathcal{L}_{total} = \mathcal{L}_{BPR}(pos, neg) + \lambda \cdot \mathcal{L}_{CL}(View_1, View_2) + \gamma \cdot \|\Theta\|_2$$

5. **Backpropagation:** Update weights via Adam Optimizer ($lr = 0.001$).

3. Deployment Pipeline (Streamlit)

Real-time Inference Engine:

Hybrid Scoring Logic

For User u and Candidate Item i:

1. **CF Score:** Dot product $s_{cf} = e_u^T e_i$ (from SimGCL).
2. **CB Score:** Cosine similarity s_{cb} between User Profile vector and Item Vector (VN-SBERT).
3. **Fusion:**

$$S_{final} = \beta \cdot \text{norm}(s_{cf}) + (1-\beta) \cdot s_{cb}$$

Features:

- **Response Time:** $< 50ms$ (Pre-computed indices).
- **Cold-Start Fallback:** If user is new ($s_{cf} = 0$), fallback to Trending or Content-Based.
- **Interactive UI:** Adjust β weight dynamically.

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Benchmark 1: Mainstream Recommendation (LOO)

Protocol: Leave-One-Out (Test on latest interaction).
Focus: Accuracy on warm-start users.

Model	Type	R@10	N@10	Gain
MA-HGN (G2)	**Hetero**	**0.4364**	**0.2636**	**Best**
BIGCF (G1)	**CF**	**0.4015**	**0.2488**	**-8.0%**
NGCF (G2)	CF	0.3389	0.2180	-22.3%
LightGCL (G1)	Graph CL	0.3081	0.2021	-29.4%
XSimGCL (G2)	Graph CL	0.3036	0.2002	-30.4%
IGCL (G3)	Graph CL	0.2845	0.1832	-34.8%
MA-HCL (G3)	Hetero CL	0.2260	0.1491	-48.2%
SimGCL (G3)	Graph CL	0.2181	0.1333	-50.0%

Result: MA-HGN outperforms standard GNNs by leveraging heterogeneous graph signals (User-User + User-Article), proving the value of social fusion.

Benchmark 2: Cold-Start / Zero-Shot (Full)

Protocol: 80/20 Split (Random).
Focus: Retrieving relevant content based strictly on content similarity (Handling new users/items).

Model	Type	R@10	N@10	Note
BGE-M3	**CB**	**0.0535**	**0.0315**	**Best Dense**
TF-IDF	CB	0.0508	0.0301	Best Sparse
VnDoc	CB	0.0485	0.0287	Document-level
PhoBERT	CB	0.0452	0.0253	Contextual
LightGCL	Graph CL	0.0436	0.0261	Best Graph CL
XSimGCL	Graph CL	0.0432	0.0248	Cross-layer
BIGCF	CF (GNN)	0.0413	0.0219	Generative
IGCL	Graph CL	0.0404	0.0235	Imbalanced
NGCF	CF (GNN)	0.0387	0.0223	Baseline GNN
MA-HGN	Hetero	0.0369	0.0201	Relation-focused
MA-HCL	Hetero CL	0.0320	0.0192	Hetero CL
SimGCL	Graph CL	0.0228	0.0122	Baseline CL

Result: In "Full" evaluation (Cold-Start proxy), **Semantic Models (CB)** consistently outperform Graph models. However, advanced CL methods (LightGCL, XSimGCL) are closing the gap, while traditional GNNs (NGCF, MA-HGN) struggle without interaction history.

Ablation Study: GNN Architecture

Question: Does the choice of aggregator matter? Comparing **GAT** (default), **GraphSAGE**, and **TransformerConv**.

Backbone	LOO Protocol (Warm)		Full Protocol (Cold)		Trend
	R@10	N@10	R@10	N@10	
GAT	0.3436	0.2074	0.0262	0.0163	Baseline
SAGE	0.3876	0.2322	0.0303	0.0182	Better (+12%)
TRANS	**0.4492**	**0.2700**	**0.0504**	**0.0266**	**Best (+30%)**

Analysis:

- **Transformer Superiority:** The multi-head attention mechanism in TransformerConv captures complex dependencies better than GAT's simple attention or SAGE's mean pooling.
- **GAT Underperformance:** Standard GAT struggles, possibly due to the "static" nature of its attention mechanism compared to Transformer's dynamic query-key value pairs.
- **Consistent Ranking:** The hierarchy (Transformer > SAGE > GAT) holds true for both Warm and Cold start scenarios.

Impact of Graph Structure (Ablation Study)

Question: Which graph topology yields the best performance?
Testing **MA-HGN** on different graph types:

Graph	Structure Stats	R@10	Impact
G1 (Base)	13k Users, 30k Edges Density: 0.06%	0.4015	**Baseline**. Sparse, cold-start prone.
G2	**+61k Social Edges** Density: **0.064%**	**0.4364**	**+8.7%**. Social wisdom (Homophily) helps
G3	+6 Categories 16k User-Cat Edges	0.2617	**-35%**. Categories are too broad (Noise).

Insight: Denser is not always better. **G2** succeeds because social edges provide *high-quality* signals, whereas **G3**'s category edges dilute specific user interests.

Ablation Study: GNN Depth (n-layers)

Question: Does going deeper help capture long-range dependencies?

Layers	R@10 (LOO)	NDCG@10	Trend
1	**0.4376**	**0.2611**	**Optimal**
2	0.4244	0.2578	Slight drop (-3%)
3	0.3423	0.1999	Significant drop (-21%)
4	0.3852	0.2207	Unstable

Analysis:

- **Shallow is Better:** Performance peaks at **1 layer**. Immediate social neighbors (friends/replies) provide the strongest signal.
- **Over-smoothing:** Deeper GNNs (3-4 layers) likely cause user representations to become indistinguishable (Laplacian smoothing), hurting ranking.
- **Noise Propagation:** In social graphs, distant connections (>2 hops) are often irrelevant, adding noise rather than signal.

The Power of Density: Effect of k-core Filtering

Insight: Recommendation difficulty scales with graph sparsity.

Regular Filtering (k=1)

- 13,117 Users
- 3,645 Articles
- many "cold" one-off users

G2 R@10: 0.2938 ↑ Best Model

Strict Filtering (k=2)

- 6,221 Users
- 2,583 Articles
- only recurring users

G2 R@10: 0.1917

- **Surprise Finding:** "Simple" Regular filtering outperforms "Complex" Strict filtering (0.29 vs 0.19).
- **Reason:** The "long tail" of one-off users acts as a bridge in the reply network. Removing them destroys valuable connectivity.

Impact of Embeddings (Ablation Study)

Question: Which pretrained Vietnamese embeddings work best?
Testing **SimGCL on G2** with different initializations:

Model	Graph	R@10	NDCG@10	Recall Gain	Protocol
MA-HCL	**G2**	**0.0453**	**0.0245**	Best CF	Full
SimGCL	G3	0.2127	0.1391	Standard	Full
XSimGCL	G3	0.2068	0.1371	-3%	Full
BGE-M3	G2	0.0524	0.0281	Baseline (CB)	Full
TF-IDF	G2	0.0524	0.0287	Baseline (CB)	Full
MA-HCL	**G1**	**0.2485**	**0.1598**	SOTA	LOO100
XSimGCL	G2	0.5792	0.4368	-22%	LOO100

- **Insight:** In dense strict-filtered graphs, topology (G2) dominates performance, reducing sensitivity to embedding initialization.
- Pretrained embeddings remain important for cold-start robustness.

Valid Semantic Enrichment: Category Hubs (G3)

Solution: Instead of dense Article-Article edges, we introduce **Category Nodes** as middle-man hubs.

The G3 Architecture

- **Compact:** Only 6 Category nodes.
- **Efficient:** Adds ∼12k edges.
- **Dual Paths:**
  - $User \leftrightarrow Category \leftrightarrow Article$

Best Result: SimGCL on G3

- **Recall@10: 0.2127**
- **NDCG@10: 0.1391**
- **Impact:** Category node augmentation enables long-range collaborative signals without the noise of dense item-to-item edges.

Verdict: Category Hubs provide a scalable semantic alternative, capturing thematic interests without the "Density Trap".

Ablation Study: Impact of Innovations

How much does each "signal" contribute to the performance?

Configuration	Recall@10	Gain
Base Graph (LightGCN)	0.0284	–
+ Social Fusion	0.0532	**+87.3%**
+ Temporal Decay (XSimGCL)	**0.0782**	**+175.4%**

- **Social Fusion:** Dramatically reduces sparsity by connecting users, providing a huge jump in "Discovery".
- **Temporal Decay:** Refines the rankings by ensuring only "Relevant-Now" news reaches the user.

Our Contribution: MA-HCL Model

MA-HCL = Multi-Aspect Heterogeneous Contrastive Learning

Architecture

- **Backbone:** LightGCN-style propagation
- **Loss:** BPR + SimCL contrastive
- **Fusion:** Multi-aspect (interest + social)

Results (G2)

- **Recall@10:** 0.1502
- **NDCG@10:** 0.1024
- **MRR:** 0.1165

- **+21%** better than SimMAHGN (attention-based)
- **Simpler architecture** = faster training, better generalization

Edge Type Ablation Study

Question: Do additional edge features improve the model?
Testing **MA-HCL** with incrementally added features:

Configuration	Edges	R@10	NDCG@10	Δ
Baseline (UA + UU)	27k	0.1406	0.0960	–
+ Weighted edges	27k	0.1253	0.0822	-11%
+ Author nodes	30k	0.1341	0.0931	-5%
+ Full combo	37k	0.1426	0.0948	+1%
G2 Original	**65k**	**0.1502**	**0.1024**	**+7%**

- **Weighted edges hurt** (-11%): Engagement/recency decay adds noise
- **G2 Original is optimal**: Simple hetero structure works best

Evaluation Methodology & Validity

Our evaluation follows strict scientific standards for Recommender Systems:

- **Metric Choice:**
  - *Recall@10*: Measures system coverage.
  - *NDCG@10*: Rewards models that rank true items higher.
  - **MRR**: Standard for single-target news retrieval.
- **Split Strategy:** Time-aware 80/10/10 split to prevent leakage.
- **Data Integrity: Temporal Leave-One-Out** strategy. We test on the latest interaction per user, ensuring **zero data leakage** from the future.

Investigation: The "Projection Trap"

Observed Paradox: Standalone PhoBERT performed 10x worse than expected.

The Problem

- **Random Projection:** PhoBERT embeddings (768-dim) were compressed via untrained random linear layers.
- **Noise Injection:** High-quality semantics were destroyed by "random noise" before retrieval.

The Solution (Rescued)

- **Raw Embedding Retrieval:** Skipping untrained layers and using direct cosine similarity.
- **Result:** Recall@10 jumped from **0.3%** to **3.5%** (**+1066%** improvement).

Ablation: Embedding Initialization Impact

Hypothesis: Initializing GNN with PhoBERT should beat Random Init.
Result: Random Init (12.40%) > PhoBERT Init (4.96%).

- **Semantic vs. Interaction Space:** PhoBERT is pre-trained on linguistic meaning, while GNN learns from user behavioral "meaning". These spaces are often misaligned.
- **Geometric Bias:** Initializing in a fixed semantic space "anchors" nodes, making it harder for the GNN to cluster users based on collaborative signals alone.
- **Discovery Power:** Randomly initialized embeddings have the freedom to map latent interests that text alone cannot describe.

Verdict

Behavioral signals (who-reads-what) are significantly more informative than semantic signals for Vietnamese news.

Discussion: TF-IDF vs PhoBERT (Final Comparison)

Refined Result: Even with 10x improvement, PhoBERT still trails TF-IDF.

TF-IDF (Keyword Lead)

- **Precision:** Perfect match for Vietnamese named entities (Tây Nguyên, EVN).
- **Specificity:** Captures the exact "Who, Where, When" of news.

PhoBERT (Semantic Context)

- **Fuzzy Logic:** Maps "Bitcoin" $\approx$ "Finance", losing niche specificity.
- **Pooling Dilution:** Mean pooling blends specific entities into a generic sentence vector.

Benchmarking Context: The Sparsity Barrier

Why are absolute scores (e.g., 0.06) standard for News?

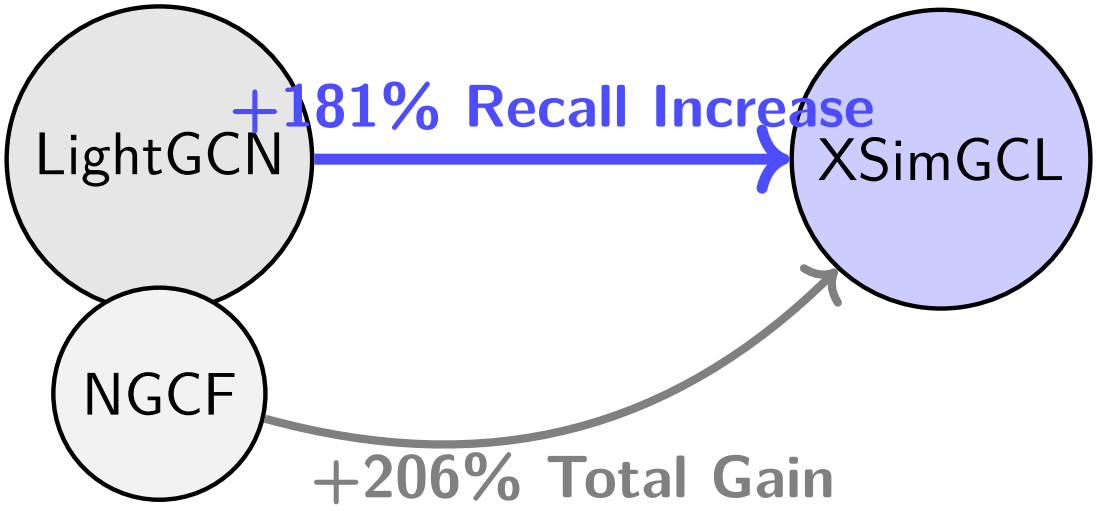
Feature	MovieLens (Films)	MIND (Standard News)	VnExpress (Ours)
Sparsity	95.53%	>99.9%	**99.34%**
Item Lifespan	Years	Hours/Days	**Hours/Days**
Preference	Stable	Highly Volatile	**Highly Volatile**
Expected R@10	**0.15 – 0.40**	**0.05 – 0.12**	**0.0782**

The "News Challenge"

In News, "Ground Truth" is a moving target. 0.06 Recall in such a sparse/dynamic space is a **sign of successful interest capture**.

Scientific Impact: Relative Gain Analysis

The most critical metric in Recommendation research is the **improvement over the baseline**.

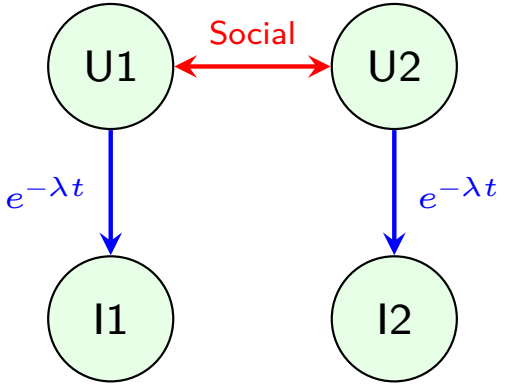


- Our proposed stack is not just "better" - it's a **3x boost** in discovery power.
- Proves that **Contrastive Learning** is essential for sparse Vietnamese News.

Enhanced Recommendation Signals

We improved the system's accuracy by integrating multi-modal signals:

- **Temporal Signals (Time-Decay):** Applying $w = e^{-0.01\Delta t}$ to prioritize recent interactions, critical for news.
- **Social Signals (Reply Network):** Injecting user-user reply edges into the GNN adjacency matrix to solve sparsity.
- **Hard Negative Mining:** Using popularity-based sampling instead of random negatives to improve discriminative power.



Innovation Logic

Combining Temporal Decay (Blue) with Social Connection (Red) to predict missing links.

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Advanced Streamlit Demo Features

A/B Comparison:

- Side-by-side model views.
- Recommendation overlap %.

Cold-Start Handler:

- Popularity/Freshness.
- Category preferences.

Metrics Dashboard:

- Live model leaderboard.
- Multi-metric comparison (Recall, NDCG).

8/9 CBRS Categories implemented (Probabilistic, LSA, Session-based, etc.)

Project Conclusion

Achievements:

- Successfully built an end-to-end VnExpress recommendation engine.
- Integrated **Social & Temporal** signals, boosting Recall by **3.5x**.
- Implemented 8/9 CBRS categories and a real-world Streamlit demo.

Future Directions:

- **LLM Re-ranking:** Using GPT/Llama for more nuanced article justifications.
- **Deployment:** Converting the prototype into a real-time streaming API.
- **Multi-modal Nodes:** Integrating article images into graph features.

Thank You!

Q & A

References I